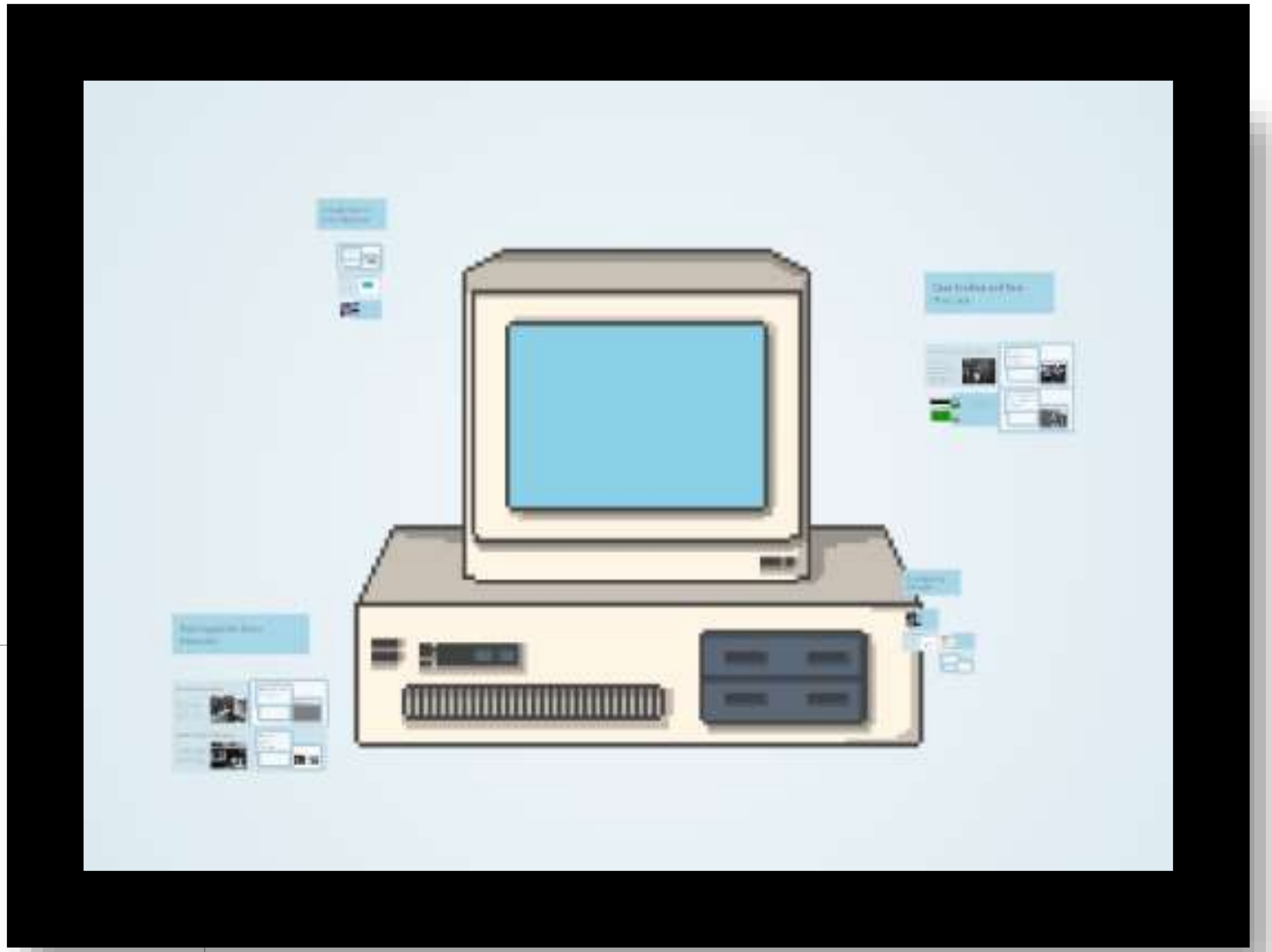
Compiler Error Detection System



CSA1402-Compiler Design

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INTRODUCTION

- A **compiler** is a software program that translates high-level programming code into machine code.
- During the compilation process, the compiler checks the source code for different types of errors before execution.
- The three major compiler errors are **Lexical Errors**, **Syntax Errors**, and **Semantic Errors**. Detecting these errors early helps developers improve code quality and reduces debugging time.
- The system analyzes the source code, identifies errors, displays recommended fixes, and generates a detailed error report.
- The project provides a simple and effective way for students and beginners to understand compiler error detection.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

- Compiler error messages can be difficult for beginners to understand.
- Identifying different types of errors manually takes time.
- A simple tool is needed to classify and explain compiler errors clearly.

Objectives

- Detect Lexical Errors.
- Detect Syntax Errors.
- Detect Semantic Errors.
- Generate a detailed error report.

TYPES OF COMPILER ERRORS

Lexical Error

- Occurs when the compiler encounters an invalid character or token.
- Example:

int @a = 10;

Reason: @ is not a valid character in a variable name.

Syntax Error

- Occurs when the program does not follow the correct programming syntax.
- Example:

int a = ;

Reason: Missing value after the assignment operator (=).

Semantic Error

- Occurs when the code is syntactically correct but logically incorrect.
- Example:

int age = "Twenty"

Reason: A string value is assigned to an integer variable, causing a type mismatch.

IMPLEMENTATION METHODOLOGY

Step 1

Designed the graphical user interface using Tkinter.

Step 2

Created a code editor to accept C source code.

Step 3

Read the program line by line.

Step 4

Applied compiler rules to detect lexical, syntax, and semantic errors.

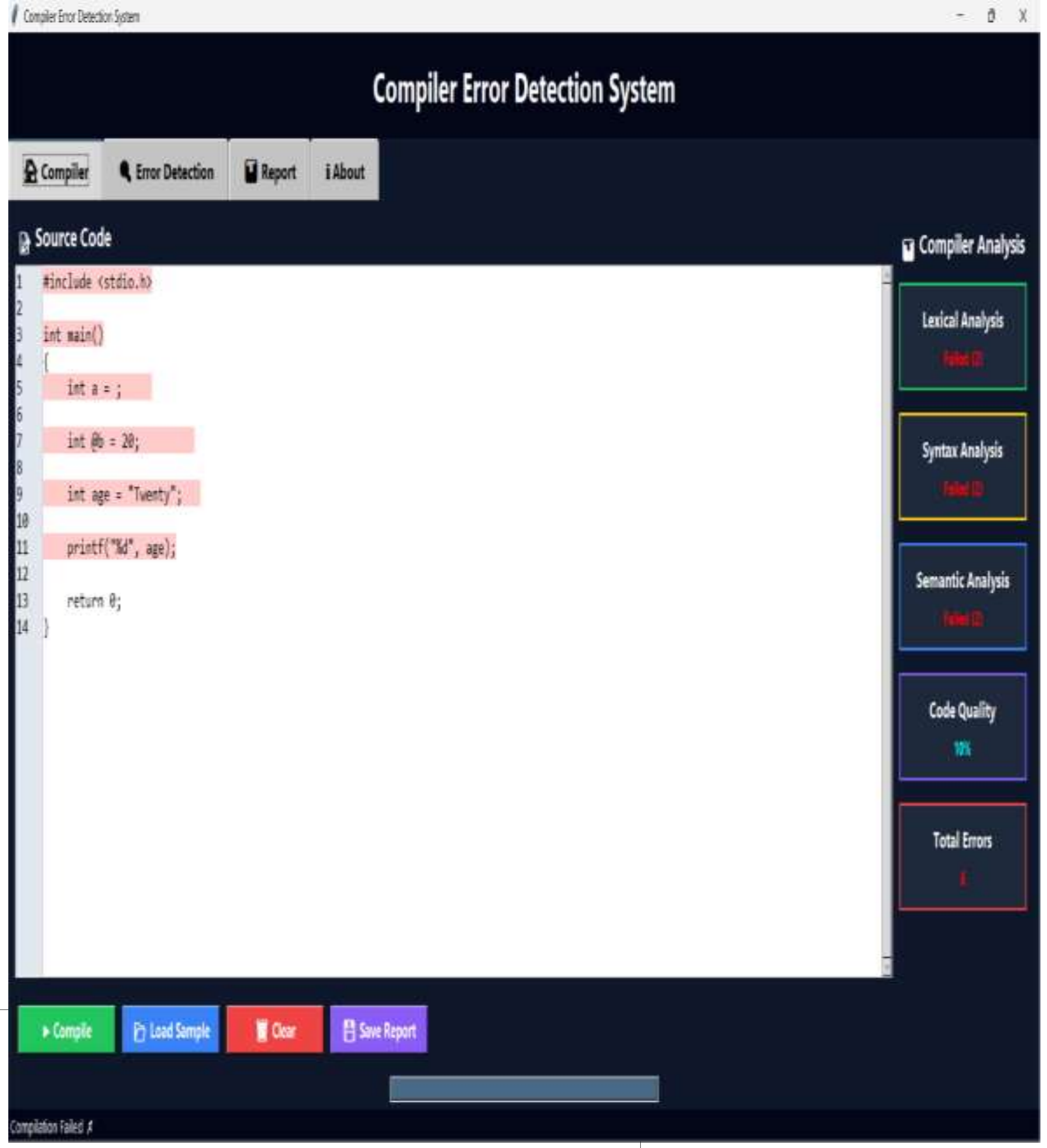
Step 5

Displayed detected errors with recommended fixes.

Step 6

Generated a compilation report.

IMPLEMENTATION RESULT

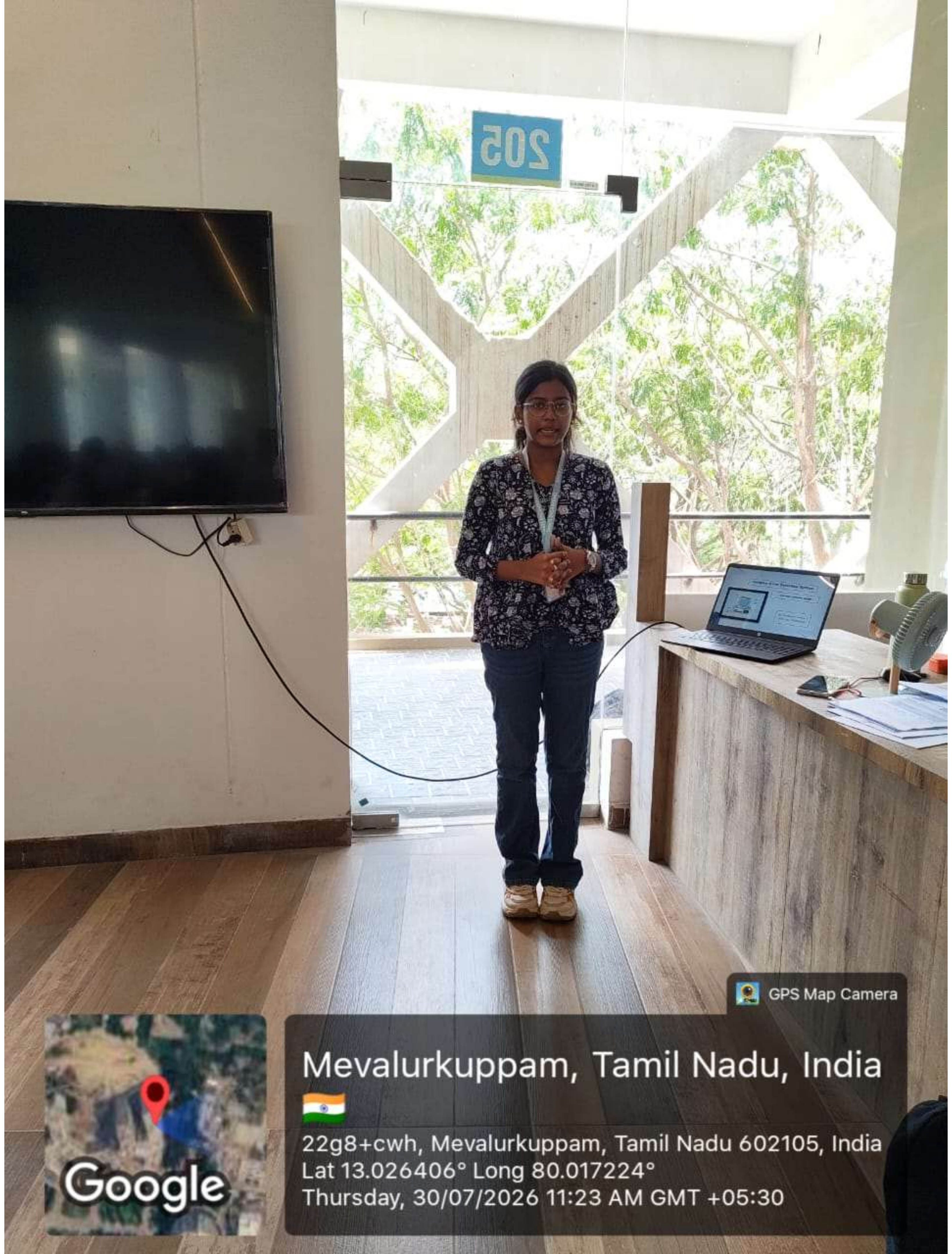


CONCLUSION

- The project successfully demonstrates the **Error Handling phase** of a compiler by identifying lexical, syntax, and semantic errors in C source code before execution.

- The application classifies compiler errors, provides recommended fixes, and generates a detailed report, making the debugging process easier and more organized.

- This project serves as an effective educational tool that helps students understand compiler error detection concepts through practical implementation and real-time analysis.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